The cardinality of the power set is 2 to the power of the cardinality

The Open Logic Project contributors

openlogic:set-theory/card-arithmetic:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-card-arithmetic/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identification of the numeral 2 with the set {0, 1} is recalled in the prose preceding the lemma (source lines 908-911) and is used to read cardexpo{2}{card{A}}; no reference inside the counted span is external.

Editorial note. The label lem:SizePowerset2Exp names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- The set 2 as {0, 1}. Same unit, the definition of omega and the identification 2 = {0, 1}. Prose before the lemma (source lines 908-911): the notation cardexpo{2}{c} relies on identifying 2 with the set {0, 1}; the identification is recalled in the running text, not inside the lemma.

Lemma 0.1. $|\wp(A)| = 2^{|A|}$, *for any* A.

Proof. For each subset $B \subseteq A$, let $\chi_B \in {}^A2$ be given by:

$$\chi_B(x) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{otherwise.} \end{cases}$$

Now let $f(B) = \chi_B$; this defines bijection $f\colon \wp(A) \to {}^A2$. So $\wp(A) \approx {}^A2$. Hence $\wp(A) \approx {}^{|A|}2$, so that $|\wp(A)| = |{}^{|A|}2| = 2^{|A|}$. □